

Yash Sinha

LinkedIn: [linkedin.com/in/yashh-sinha2828](https://www.linkedin.com/in/yashh-sinha2828)
GitHub: github.com/Yashh2828

Email: yashhsinha2809@gmail.com
Mobile: +91-8299582905

Summary

Computer Science undergraduate with specialization in AI/ML and hands-on experience in building AI-powered applications and backend systems using Python, Flask, and MongoDB. Experienced in developing scalable systems, workflows, and real-time dashboards. Published a research paper on an AQI prediction model integrated with a travel website.

Skills

- **Languages:** Python
- **Concepts:** OOP, DBMS, REST APIs, Authentication, RBAC
- **Frameworks:** FastAPI, Flask, TensorFlow, Bootstrap
- **AI/ML:** RAG, Transformers, FAISS, scikit-learn, Pandas, NumPy, PyTorch, Prompt Engineering
- **Databases:** MongoDB Atlas, MySQL
- **Tools:** Git, GitHub, VS Code, Google Colab, PyCharm, Windsurf, UiPath

Experience

Hexaware Technologies

Feb 2026 – May 2026

- Applied Python and OOP principles for data preprocessing, analysis, and machine learning workflows using NumPy and Pandas.
- Worked with Generative AI concepts including LLMs, prompt engineering, and RAG pipelines.
- Built automation workflows using UiPath including RE Framework, queues, Orchestrator, and data scraping.
- Developed automation solutions integrating Excel, PDF, and email workflows.

Uttar Pradesh Metro Rail Corporation (UPMRC), Lucknow

May 2025 – June 2025

- Developed an internal ticketing system for equipment fault tracking and issue management.
- Built RESTful APIs and analytics dashboards, reducing manual reporting effort by 60%.
- Integrated MongoDB with optimized indexing for efficient data retrieval.
- Designed real-time dashboards for ticket monitoring and analytics.

Projects

Insurance Policy Explainer

AI + Full-Stack Web App

- Built a RAG-based insurance analysis system using FastAPI, FAISS vector search, and transformer embeddings for contextual clause retrieval.
- Implemented LLM-based clause analysis to generate policy explanations and detect high-risk terms.
- Reduced manual policy review effort by approximately 90%.

Equipment Fault Ticketing System

Flask + Frontend + MongoDB

- Built RESTful APIs and analytics dashboards, reducing manual reporting effort by 60%.
- Developed RBAC-enabled dashboards for End Users, Admins, and Super Admins.
- Implemented a complete ticket lifecycle with state transitions and authentication and authorization workflows.

Air Quality Index (AQI) Prediction System

Machine Learning

- Built predictive machine learning models using TensorFlow and scikit-learn for AQI forecasting.
- Trained the model on 5 years of CPCB AQI data and achieved 85% prediction accuracy.
- Visualized predicted versus actual AQI trends for performance evaluation.

Certifications

- Amazon Summer ML School — Amazon — 2025
- Prompt Engineering for LLMs — OpenAI — 2025
- OCI AI Foundations Associate — Oracle — 2025
- GitHub Foundations — GitHub — 2025
- Applied Machine Learning — Reliance — 2026

Education

2022–2026	B.Tech in Computer Science and Engineering, Amity University, Lucknow
2022	Intermediate (Physics, Chemistry, Mathematics), Seth M.R. Jaipuria School, Lucknow
2020	High School, Seth M.R. Jaipuria School, Lucknow